PHATE, Mutual kNN, and Intersection kNN Graph Constructions

gflow development note

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Abstract

This note gives a self-contained description of three graph constructions on a finite data set: the PHATE adaptive affinity graph, the mutual k-nearest-neighbor graph, and the intersection k-nearest-neighbor graph. The main point is that the PHATE graph with additive symmetrization has an adaptive-radius edge condition

$$d_{ij} \le r_\tau \max(\sigma_i, \sigma_j),$$

where $r_\tau = (-\log \tau)^{1/a}$. In contrast, the mutual kNN graph has the smaller radius condition

$$d_{ij} \le \min(\sigma_i, \sigma_j)$$

when there are no nearest-neighbor ties. The intersection kNN graph is not a radius graph: it connects points whose k-neighbor sets have nonempty intersection. Thus mutual kNN is naturally contained in the PHATE support under matching conventions, but PHATE and intersection kNN are not generally nested. Figures illustrate these distinctions on roots of unity, a larger uniform circle, and noisy circles constructed with a sample Fermat distance.

1 Setup and conventions

Let

$$X = \{x_1, \dots, x_n\} \subset \mathbb{R}^p$$

be a finite data set, and let $D = (d_{ij})$ be a symmetric nonnegative distance matrix on X, with $d_{ii} = 0$. In Euclidean examples,

$$d_{ij} = \|x_i - x_j\|_2.$$

In the noisy-circle examples below, D is instead a sample Fermat distance computed from the observed points.

For a fixed integer $k \in \{1, \dots, n-1\}$, define $N_k(i)$ to be the set of the k nearest neighbors of x_i, excluding x_i itself. Unless otherwise noted, ties are resolved by vertex index. This tie convention is only needed for finite examples with exact symmetry; the formulae below are cleanest when there are no ties.

Define the local k-neighbor radius

$$\sigma_i = d_{i,(k)},$$

where $d_{i,(k)}$ is the distance from x_i to its k-th non-self nearest neighbor. Equivalently, in the no-tie case,

$$j \in N_k(i) \iff d_{ij} \le \sigma_i.$$

2 Mutual kNN graph

The mutual k-nearest-neighbor graph, denoted $G_{\mathrm{mKNN}} = (V, E_{\mathrm{mKNN}})$, has vertex set $V = \{1, \ldots, n\}$. An undirected edge $\{i, j\}$, $i \neq j$, is present if and only if

$$j \in N_k(i) \quad \text{and} \quad i \in N_k(j).$$

When there are no ties, this is equivalent to

$$d_{ij} \leq \sigma_i \quad \text{and} \quad d_{ij} \leq \sigma_j,$$

or

$$d_{ij} \leq \min(\sigma_i, \sigma_j).$$

Thus mKNN is a conservative reciprocal-neighborhood graph. It keeps edges only when both endpoints regard each other as locally close.

3 Intersection kNN graph

The intersection k-nearest-neighbor graph, denoted $G_{\mathrm{ikNN}} = (V, E_{\mathrm{ikNN}})$, has an edge $\{i, j\}$ if the two k-neighborhoods share at least one point:

$$\{i, j\} \in E_{\mathrm{ikNN}} \quad \Longleftrightarrow \quad N_k(i) \cap N_k(j) \neq \varnothing.$$

This is not a direct radius condition on d_{ij}. Two points can be connected because they both regard a third point as nearby, even if i and j are not mutually close. Consequently, G_{ikNN} can be much denser than mKNN or PHATE, but it need not contain PHATE in every finite configuration.

In gflow's native ikNN implementation, the graph also carries an edge weight. For an ikNN edge $\{i, j\}$, the stored native length is an intersection-mediated detour length of the form

$$\ell_{ij}^{\mathrm{ikNN}} = \min_{c \in N_k(i) \cap N_k(j)} \{d_{ic} + d_{jc}\}.$$

This is a length, not a similarity, but it is not usually the direct chord length d_{ij}.

4 PHATE adaptive affinity graph

PHATE constructs a directed adaptive affinity before symmetrization. For each point i, let

$$\sigma_i = b\, d_{i,(k)}$$

where $b > 0$ is a bandwidth scale. In the examples below $b = 1$. For decay parameter $a > 0$, define

$$A_{ij} = \exp\left[-\left(\frac{d_{ij}}{\sigma_i}\right)^a\right], \qquad A_{ii} = 1.$$

PHATE then thresholds small directed affinities:

$$A_{ij} = 0 \quad \text{if} \quad A_{ij} < \tau,$$

where $\tau = 10^{-4}$ is the default threshold used in these examples.

The directed threshold condition is

$$\exp\left[-\left(\frac{d_{ij}}{\sigma_i}\right)^a\right] \geq \tau.$$

Taking logarithms gives

$$d_{ij} \leq r_\tau \sigma_i, \qquad r_\tau = (-\log \tau)^{1/a}.$$

For $a = 40$ and $\tau = 10^{-4}$,

$$r_\tau = (-\log 10^{-4})^{1/40} \approx 1.057.$$

So in one directed row, PHATE keeps nearly the same radius as the k-nearest-neighbor radius when $a = 40$. Smaller a produces a larger effective threshold radius.

PHATE then symmetrizes the directed kernel. With additive symmetrization,

$$K_{ij} = \frac{A_{ij} + A_{ji}}{2}.$$

The undirected PHATE support is

$$E_{\text{PHATE}} = \{\{i,j\} : i < j,\ K_{ij} > 0\}.$$

Because $K_{ij} > 0$ if and only if at least one directed affinity survives thresholding,

$$\{i,j\} \in E_{\text{PHATE}} \iff d_{ij} \leq r_\tau \sigma_i \quad \text{or} \quad d_{ij} \leq r_\tau \sigma_j.$$

Equivalently,

$$\{i,j\} \in E_{\text{PHATE}} \iff d_{ij} \leq r_\tau \max(\sigma_i, \sigma_j).$$

If both directed affinities survive, K_{ij} is their average. If only one direction survives, K_{ij} is half of the surviving directed affinity.

Finally, PHATE row-normalizes K to obtain a Markov diffusion operator

$$P_{ij} = \frac{K_{ij}}{\sum_{\ell=1}^{n} K_{i\ell}}.$$

The graph support discussed in this note is the nonzero off-diagonal support of K, not the row-normalized matrix P.

5 Relations among the graph supports

Assume that all three constructions use the same distance matrix D, the same k, bandwidth scale $b = 1$, additive PHATE symmetrization, and no nearest-neighbor ties. Then

$$\{i,j\} \in E_{\text{mKNN}} \iff d_{ij} \leq \min(\sigma_i, \sigma_j).$$

Since

$$\min(\sigma_i, \sigma_j) \leq r_\tau \max(\sigma_i, \sigma_j)$$

for $r_\tau \geq 1$, we have

$$E_{\text{mKNN}} \subseteq E_{\text{PHATE}}.$$

This inclusion is the precise sense in which PHATE is less restrictive than mKNN.

However, E_{ikNN} is not governed by a direct radius threshold. It is governed by the set condition

$$N_k(i) \cap N_k(j) \neq \varnothing.$$

Therefore one should not write a general nesting relation such as

$$E_{\text{PHATE}} \subseteq E_{\text{ikNN}} \quad \text{or} \quad E_{\text{ikNN}} \subseteq E_{\text{PHATE}}.$$

In many sampled manifolds, ikNN has more edges than PHATE because many point pairs share neighbors. But this is an empirical edge-count tendency, not a theorem of set inclusion.

6 Root-of-unity examples

This section shows deterministic finite examples on the unit circle. The points are

$$x_m = (\cos(2\pi m/n), \sin(2\pi m/n)), \qquad m = 0, \ldots, n-1.$$

For $n = 3, 4, 5$, we use $k = 2$, $a = 40$, and $\tau = 10^{-4}$. For $n = 100$, we use $k = 10$. All graph constructions use Euclidean distances.

6.1 Three roots of unity

For $n = 3$ and $k = 2$, every point has the other two points as its two nearest neighbors. Therefore

$$N_2(i) = \{1, 2, 3\} \setminus \{i\}.$$

Every pair is mutually 2-nearest, so mKNN is the complete graph K_3. Since every distance equals σ_i, every pair also satisfies

$$d_{ij} \leq r_\tau \sigma_i$$

in both directions, so PHATE is also K_3. Finally, any two distinct vertices have neighbor sets with nonempty intersection, so ikNN is K_3 as well.

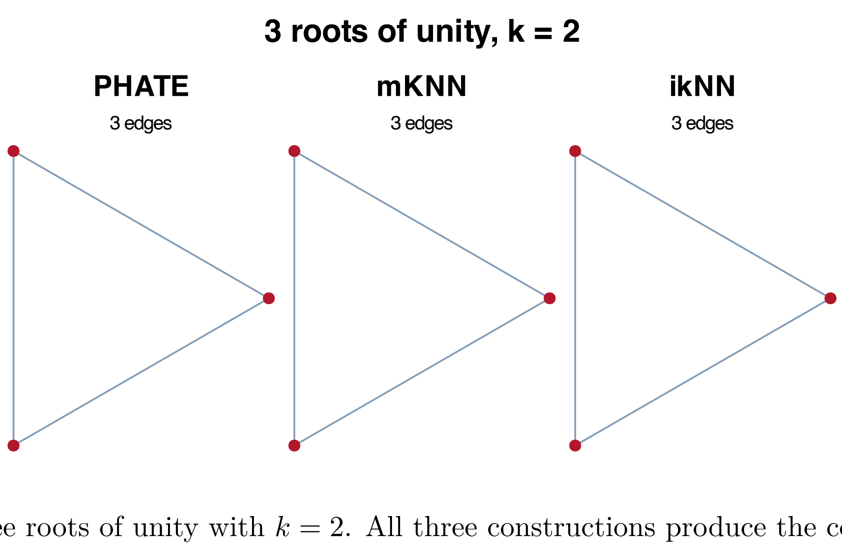

Figure 1: Three roots of unity with $k = 2$. All three constructions produce the complete graph.

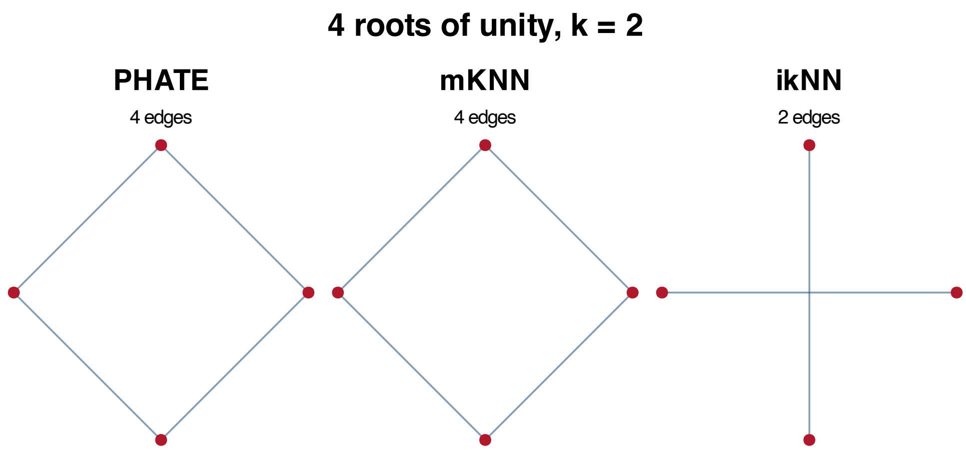

Figure 2: Four roots of unity with $k = 2$. PHATE and mKNN produce the cycle, while ikNN connects opposite vertices through shared nearest neighbors.

6.2 Four and five roots of unity

For $n = 4$ and $k = 2$, mKNN and PHATE both recover the cycle edges. The ikNN graph, however, connects points with common nearest neighbors; in the square this produces the two diagonals. This example is useful because it shows that ikNN is not simply a superset of PHATE.

For $n = 5$ and $k = 2$, PHATE and mKNN again recover the adjacent cycle. The ikNN graph connects points at graph distance two around the circle, forming a pentagram-like support.

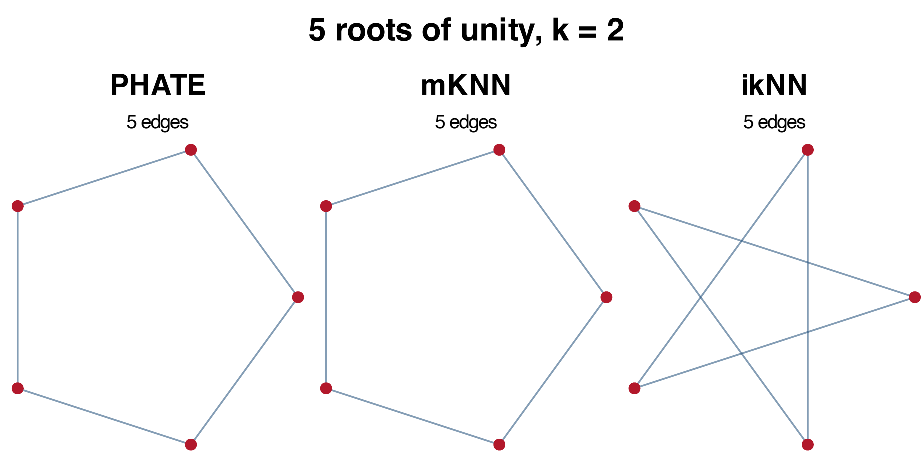

Figure 3: Five roots of unity with $k = 2$. PHATE and mKNN keep adjacent circle edges; ikNN connects pairs that share a nearest neighbor.

6.3 One hundred roots of unity

For $n = 100$ and $k = 10$, PHATE remains close to a local circular band because $r_\tau \approx 1.057$. mKNN is the reciprocal-neighborhood graph. ikNN is visibly denser because many pairs share at least one of their ten nearest neighbors.

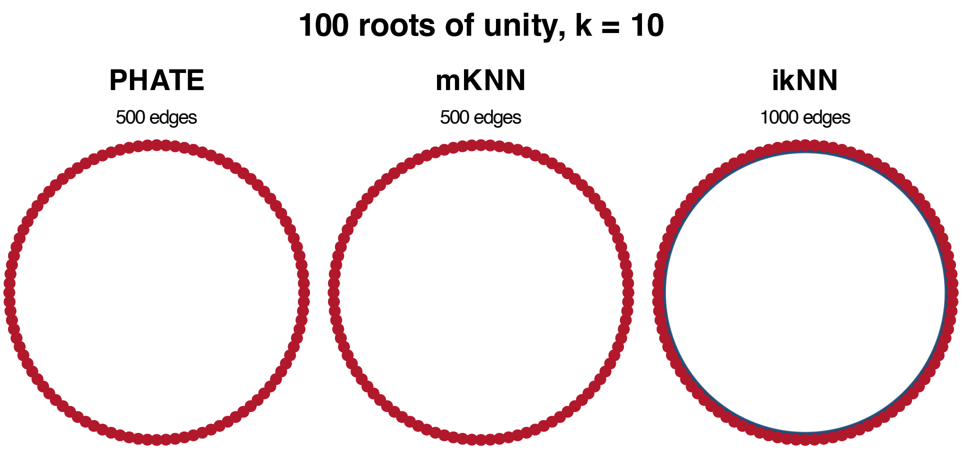

Figure 4: One hundred roots of unity with $k = 10$. ikNN is denser because it is based on shared neighborhoods rather than reciprocal or adaptive-radius proximity.

7 Noisy circle with sample Fermat distance

The term *Fermat distance* is standard in the density-aware metric learning literature. For a sample X, a common discrete version is a shortest-path metric whose edge costs are powers of Euclidean lengths. In the examples here, we first build a base kNN graph on the observed noisy circle and assign each base edge $\{i, j\}$ the cost

$$c_{ij} = \|x_i - x_j\|_2^\alpha,$$

with $\alpha = 2$. The sample Fermat distance is then

$$d_F(i, j) = \min_{\gamma: i \rightsquigarrow j} \sum_{\{u,v\} \in \gamma} c_{uv}.$$

This is a graph-based sample version of Fermat-type distances: when $\alpha > 1$, long single jumps are penalized more heavily, and shortest paths tend to follow chains of nearby sample points. We then construct PHATE, mKNN, and ikNN graphs using d_F as the input distance matrix. The figures are drawn in the original noisy Euclidean coordinates so that the supports can be inspected visually.

8 Summary

The three graph constructions answer different local questions:

- mKNN asks whether two points are reciprocal nearest neighbors:

$$d_{ij} \leq \min(\sigma_i, \sigma_j).$$

- PHATE asks whether at least one endpoint sees the other within its adaptive threshold radius:

$$d_{ij} \leq r_\tau \max(\sigma_i, \sigma_j).$$

- ikNN asks whether two points share at least one neighbor:

$$N_k(i) \cap N_k(j) \neq \varnothing.$$

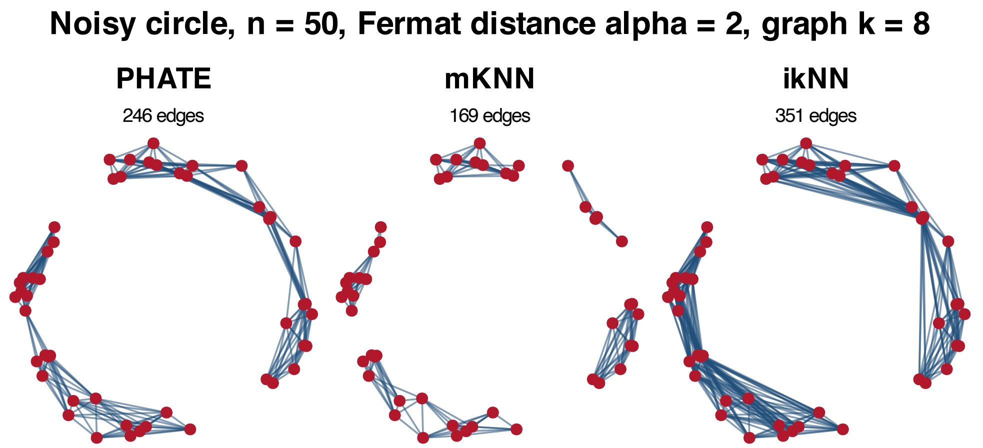

Figure 5: Noisy circle, $n = 50$, graph $k = 8$, sample Fermat distance with $\alpha = 2$.

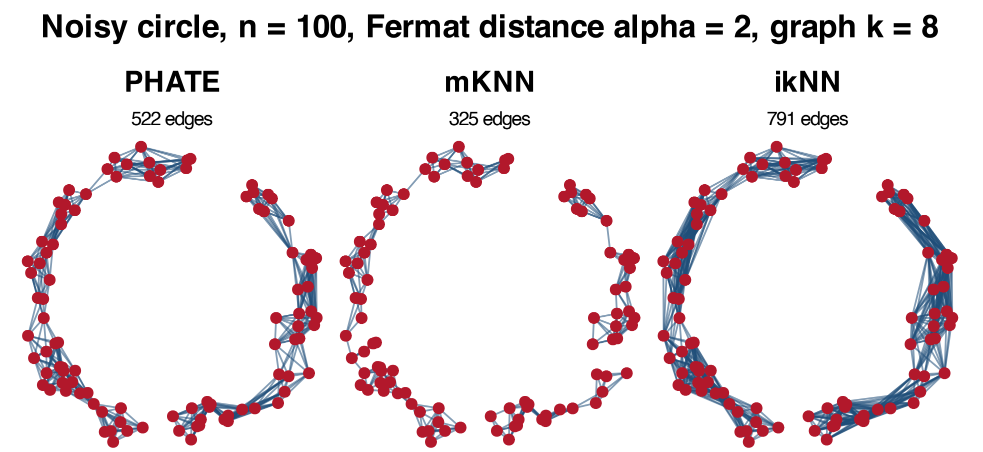

Figure 6: Noisy circle, $n = 100$, graph $k = 8$, sample Fermat distance with $\alpha = 2$.

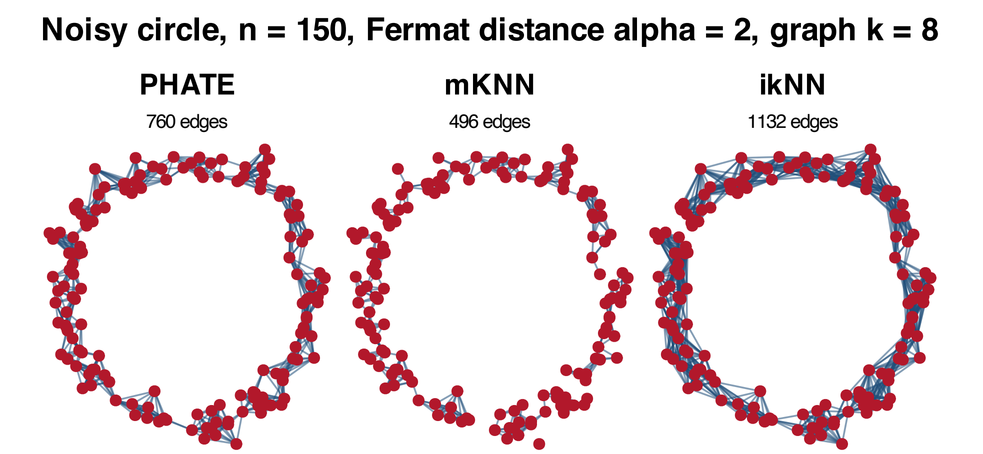

Figure 7: Noisy circle, $n = 150$, graph $k = 8$, sample Fermat distance with $\alpha = 2$.

Thus mKNN is contained in PHATE under the stated assumptions, while ikNN is not generally nested with PHATE. In many data examples ikNN is denser, but the roots-of-unity examples show that the relationship is not merely an edge-count hierarchy. This distinction matters when the goal is geodesic reconstruction: PHATE support may improve local geometric connectivity even when its support is sparser than ikNN, while ikNN may introduce shared-neighborhood edges that are not direct adaptive-radius edges.

References

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